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Preprint *in* Ultramicroscopy · February 2019

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Design of a charged particle beam phase plate for transmission electron microscopy

Philip J. B. Koeck
Royal Institute of Technology,
School of Engineering Sciences in Chemistry, Biotechnology and Health
and Karolinska Institutet, Department of Biosciences and Nutrition, Huddinge, Sweden
e-mail: Philip.Koeck@ki.se

Keywords: phase contrast, electrostatic phase plate, weak phase object approximation, transfer function, transmission electron microscopy, charged particle beam

Acknowledgements: I acknowledge Dimitri Schritt for coming up with the original idea and Fred Sigworth for sharing and help with his Matlab code. I also thank Marco Beleggia and Marek Malac for helpful discussions.

Abbreviations:
TEM, transmission electron microscopy/microscope
PCTF, phase contrast transfer function

Introduction

Zernike phase contrast has been recognized as a means of recording high resolution images with high contrast using a transmission electron microscope (Danev and Nagayama 2001). This imaging mode can be used to image typical phase objects such as unstained biological molecules or cryo sections of biological tissue. According to the original proposal reviewed in (Edgcombe 2017) and references therein the Zernike phase plate applies a phase shift of $\pi/2$ to all scattered electron beams outside a given scattering angle and an image is recorded at Gaussian focus or slight under-focus (between Gaussian focus and Scherzer defocus). Alternatively a phase shift of $-\pi/2$ is applied to the central beam using a Boersch-style (Schultheiss et al. 2010) or a hole-free phase plate (Malac et al. 2012), (Danev et al. 2014). The resulting image will have an almost perfect phase contrast transfer function (PCTF) (close to 1) from a given lowest spatial frequency up to a maximum resolution determined by the wave length, the amount of defocus and the spherical aberration of the microscope. All phase plates in use so far suffer from similar problems due to charging of solid parts that are hit by the electron beam (Glaeser 2013; Glaeser et al. 2013). I present a suggestion for an electrostatic phase plate that is completely free of material parts that can charge in an uncontrolled way, since the charge carrier consists of focused beams of negatively charged particles, such as the electron beam used in an e-beam lithography instrument, located in the back focal plane or equivalent at right angles to the optical axis of the microscope. A charged particle beam or line charge essentially adds a roughly linear term to the lens aberration function $\chi(u)$, as shown in equation (5), and thus considerably improves the low resolution contrast transfer, even if it falls a bit short of the ideal phase plate originally envisioned. I

present simulations for a design with two such particle beams and give estimates of the beam current and charge density required. I also discuss the possibility of using more than two beams or even just a single one. Other suggestions and designs for solid-matter-free phase plates or phase plates whose solid parts should not interfere with the TEM-beam can be found in (Glaeser 2013) and (Tavabi et al. 2018).

Methods

The projected electrostatic potential due to the phase plate was calculated from the charge density distribution using the Poisson equation, which relates the electrostatic potential v(x, y, z) to the charge density ρ(x, y, z).

$$\Delta\, v(x,y,z) = -\frac{1}{\varepsilon_0}\,\rho(x,y,z) \qquad \text{equation (1)}$$

Using a theorem for the Fourier transform of the second derivative in three dimensions (Laplace operator) of a function we can write this expression in Fourier space as:

$$4\,\pi^2\left(u_x^2+u_y^2+u_z^2\right)V\left(u_x,u_y,u_z\right)=\frac{1}{\varepsilon_0}P\left(u_x,u_y,u_z\right) \qquad \text{equation (2)}$$

In equation (2) upper case letters stand for functions in Fourier space, whereas the corresponding lower case letters in equation (1) stand for the corresponding functions in real space.

By choosing $u_z = 0$ we therefore get a relation in Fourier space between the projected charge density and the projected electrostatic potential:

$$4\,\pi^2\left(u_x^2+u_y^2\right)V\left(u_x,u_y,0\right)=\frac{1}{\varepsilon_0}P\left(u_x,u_y,0\right) \qquad \text{equation (3)}$$

Clearly some care has to be taken to avoid dividing by zero at the origin of Fourier space when calculating the projected potential from the projected charge density since it leads to an undefined or infinite constant offset for the potential depending on whether the charge density has a zero or non-zero constant term, respectively. Obviously only relative potential values are possible in the end. I calculated the projected potential produced by one, two and three beams of negatively charged particles using the above relation.
The projected electrostatic potential was calculated for an area of 200 by 200 μm with a pixel size of 0.5 μm. The edge of this area (100 μm from the optical axis) corresponds to 10 nm^{-1} spatial frequency in the imaging simulations assuming that the phase plate is positioned in the back focal plane of a 200 kV TEM with a focal length of 4 mm.

Imaging simulations were carried out on the projection of a phantom representing a small protein hexamer in top view.
The electrostatic potential distribution inside trypsin inhibitor (Protein data Bank entry 4pti), a small protein with a diameter of about 2.5 - 3 nm, was calculated using Matlab-code written by Shang and Sigworth, which treats the molecule as a collection of neutral atoms as

described in (Shang and Sigworth 2012). In this program each atom is assigned an atomic radius and a single potential value, obtained at the limit of low scattering angles from the parametrization of (Kirkland 2010).
All further steps of phantom construction and electron microscopic imaging simulation were implemented in Khoros ((Konstantinides and Rasure 1994)). All plots were produced using Origin™ 7.5 (www.originlab.com).
The phantoms used as specimens for image simulation were generated by arranging 6 copies of the electrostatic potential distribution in an oligomer with C6 symmetry.
All imaging simulations were done for a 200 kV microscope with a C_s of 2 mm. For this acceleration voltage the electron wave length becomes 0.0025 nm and the interaction constant σ is 0.0073 rad / V nm. The voxel size of the phantom and the pixel size of the projection and images are 0.5 Å, giving a Nyquist limit of 1 $Å^{-1}$.
When the electron wave travels down the column and hits the specimen it first enters a layer of vitrified water, which is modelled as a constant electrostatic potential distribution of about 4.9 V for all voxels occupied by water. Since only the contrast difference between protein and water is of interest, the constant value of 4.9 V was subtracted from the phantom before calculating a projection. In the vicinity of the protein the potential drops below this constant value to take into account the finite distance between water molecules and the atoms forming the surface of the protein. For details see (Shang and Sigworth 2012).
The upper and lower surfaces of the water layer are assumed to be flat and orthogonal to the direction of the electron beam (z-axis).
The exit wave was calculated from the projected electrostatic potential $proj(\vec{x})$ of the protein phantom using equation (4) without approximating for weak phase objects. The position vector $\vec{x}$ in the following equations is two-dimensional.

$$\Psi_{ex}(\vec{x}) = e^{i\,\sigma\,proj(\vec{x})} \qquad \text{equation (4)}$$

The specimen was treated as a single projection of the electrostatic potential of the protein since for the considered specimen thickness a multislice simulation gives results that are almost indistinguishable from simulations based on the projection assumption (Koeck and Karshikoff 2015). To account for amplitude contrast using the homogenous object assumption I added an imaginary term to the projection, which is proportional to the real part with a relative amplitude of 0.07.
The Fourier transform of the exit wave was then modified by the lens aberration function χ(u) and by a phase factor describing the effect of the phase plate as shown in equation (5) where pp(u) stands for the projected electrostatic potential produced by the phase plate calculated by Fourier back transforming $P(u_x, u_y)$ from equation (3) and applying it to the diffraction plane of the electron microscope by regarding it as a function of spatial frequency at the specimen level.

$$\Psi_{pc}(\vec{u}) = \Psi_{ex}(\vec{u})\; e^{-i\,\chi(\vec{u})}\; e^{-i\,\sigma\,pp(\vec{u})} \qquad \text{equation (5)}$$

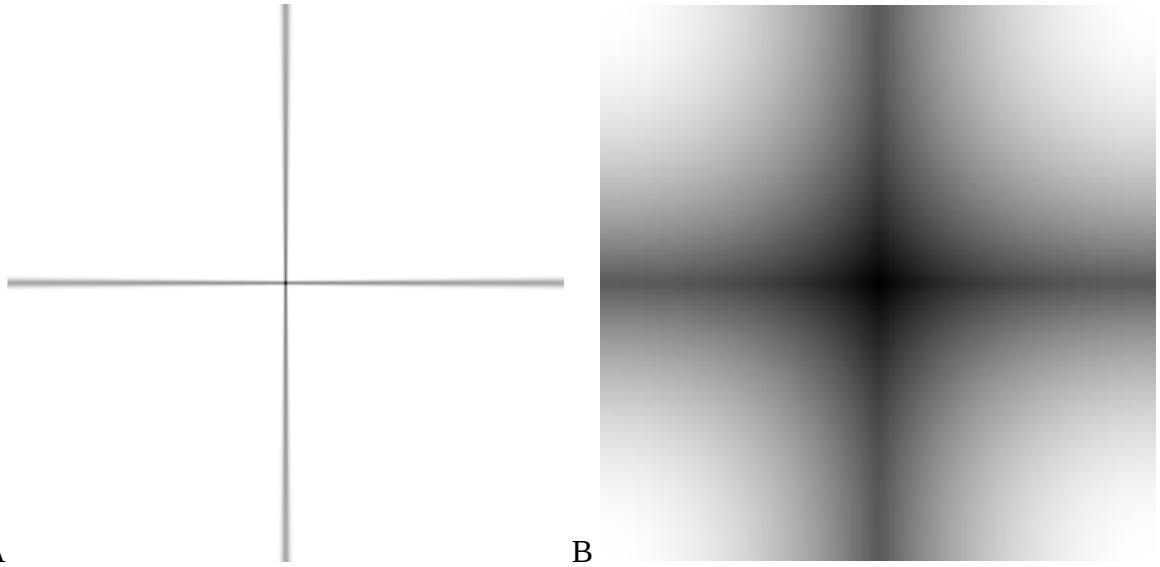
Partial spatial coherence of the electron source was taken into account by multiplying with a Gaussian envelope with a standard deviation of 0.25 Nyquist in Fourier space. To arrive at an

image, the absolute square of the result of equation (5) was taken after transforming back into real space.

Results

Dual beam

A phase plate consisting of two Gaussian beams of negatively charged particles is shown in projection in figure 1A, the corresponding projected electrostatic potential is shown in figure 1B.



A B

Fig. 1: A: Projected charge density of the electron beams (black: -5.9 x 10^{-9} pC/μm^2, white: 0 pC/μm^2). B: projected electrostatic potential (black: 0 V nm, white: 1738 V nm). The area shown in the 2D-plots above is 200 by 200 μm and the pixel size used is 0.5 μm. The edge of the plots corresponds to 10 nm^{-1} spatial frequency in the imaging simulations assuming that the phase plate is positioned in the back focal plane of a 200 kV TEM with a focal length of 4 mm. The convergence angle of each beam is about 0.02 rad and the smallest beam diameter roughly 1 μm.

In the dual beam simulation the linear charge density of one beam is about 3x10^{-7} pC/μm. This means the linear particle density for singly charged particles is about 2 particles/μm. This result retroactively justifies modelling the phase plate beam as a continuous charge distribution, given that the pixel size used in the calculations is 0.5 μm.
Assuming a speed of about 20 000 000 m/s, which is the speed of 1 keV electrons, the above charge density gives a beam current of about 5 μA.
According to table 7.2 in (Adams and Barbante 2015) and also (after correction of the unit for current density from A/m^2 to A/cm^2) table 5.1 in (Williams and Carter 2009) the current that can be achieved with a LaB_6 source at 100 keV is about 100 μA. For a kinetic energy of 1 keV the current is therefore about 10 μA. Assuming that such a beam can be focused sufficiently it seems likely that an electron beam with the required charge density can be produced with existing technology.
The above values have all been estimated assuming that the phase plate is located in the back focal plane. If it is located in another focal plane, such as in the projector system, where the

diffraction pattern is magnified, the required linear charge density is reduced (by the magnification factor) and, in addition, the beam doesn't need to be focused so strongly. Furthermore, the phase plate beams can have a relatively large convergence angle so that charge density is only high close to the center of the focal plane. In fact this is even an advantage for its performance as a phase plate. Another way of reducing charge density is to give the beams a strongly elliptical (astigmatic) profile, stretched in the direction of the TEM's optical axis since only the projected potential matters for the phase shift.

Another important question is how stable the phase plate beams will be when they are exposed to the primary TEM-beam.
An entirely classical estimate is presented in the following.
The velocity of the 1 keV phase plate electrons is about 20 000 000 m/s, whereas the velocity of the 200 keV TEM-electrons is about 200 000 000 m/s. This means that while a TEM-electron has traveled 2 mm a phase plate electron has traveled 200 μm and has completely traversed the relevant area of the focal plane. The TEM electrons thus encounter an apparently continuous beam of charge as they approach the focal plane, and it should be sufficient to model the phase plate electrons as a continuous charge distribution. Additionally the projected potential generated by point charges is a slowly changing function which increases the smoothness of the projected potential generated by the phase plate beams.
The TEM-electrons spend an equal amount of time above and below the focal plane, which means that for most TEM-electrons the change in momentum due to interacting with the phase plate electrons is close to zero once they have left the vicinity of the focal plane. The exception to this occurs if a TEM-electron actually collides with a phase plate electron and completely knocks it out of the beam.
Clearly, however, the phase plate beams will broaden somewhat due to interaction with the TEM-electrons. To get an estimate of this broadening I use a simple mechanical model with an initially stationary phase plate electron and a TEM-electron coming from infinity, passing the phase plate electron at a small distance d (1 nm for example) and proceeding in a straight line after the close encounter.
The acceleration of the phase plate electron in z-direction due to the repulsive force of the approaching TEM-electron is given by

$$a = \frac{1}{4\pi\varepsilon_0 m_e}\frac{e^2}{z^2 + d^2}$$

Here the z-axis is the optical axis of the TEM, d is the closest distance between the TEM-electron and the phase plate electron, e is the charge and m_e is the mass of an electron. The TEM-electron approaches from $z = -\infty$ and the focal plane is positioned at $z = 0$. I'm assuming that most of the acceleration of the phase plate electron is in the z-direction. The acceleration in the direction of the phase plate beam is irrelevant since it doesn't affect the charge density on average. The acceleration in the direction orthogonal to both beams is very small since it only becomes important when the TEM-electron is very close to the phase plate electron (for z comparable to d).
The maximum velocity of the phase plate electron in z-direction becomes

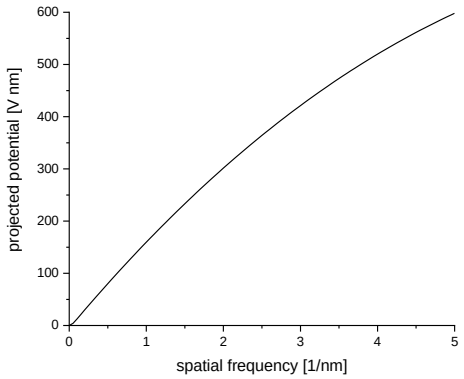
$$v = \int a\, dt = \int a\frac{dz}{v_e} = \frac{e^2}{4\pi\varepsilon_0 m_e v_e}\int_{-\infty}^{0}\frac{dz}{z^2 + d^2}$$

with v_e being the velocity of the TEM-electron, which is approximately constant.
With the indefinite integral $\int \frac{dz}{z^2+d^2} = \frac{1}{d} atan \frac{z}{d}$, the maximum velocity becomes

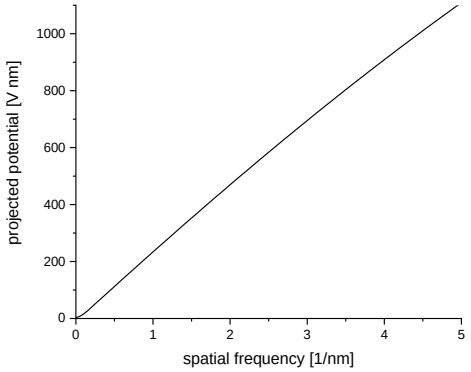
$$v = \frac{e^2}{8\varepsilon_0 m_e v_e d}$$

With the following values: e = 1.6 x 10^{-19} C, ε_0 = 8.85 x 10^{-12} $C^2/N\ m^2$, m_e = 9.1 x 10^{-31} kg, v_e = 2 x 10^8 m/s and d = 10^{-9} m (1 nm) we get a maximum velocity in z-direction of 2000 m/s for the phase plate electrons. This is only 10^{-4} times the velocity of the phase plate electrons in the direction of the phase plate beam. This means that during the time that the phase plate electrons travel 100 μm across the focal plane their path deviates by less than 10 nm from the intended and the beam broadening due to temporary momentum transfer is in that size range. It seems likely that beam broadening due to repulsion between neighboring phase plate electrons is larger than the above.

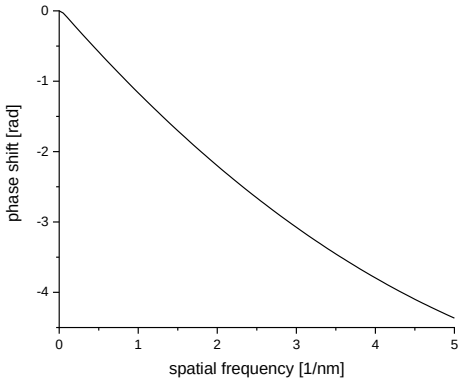
In figure 2A and B I show plots of the projected electrostatic potential in V nm. Figure 2A shows the potential along a particle beam, whereas figure 2B shows the potential diagonally between the two beams. Note that the two beams don't need to actually intersect, but can pass each other closely, since only the projected potential is of interest.
A projected potential of 200 V nm leads to a phase shift of about π/2 for 200 keV electrons, the same as the phase shift due to 20 nm of carbon film. In figure 2C and D I show plots of the relative phase shifts due to the phase plate again along a beam in C and diagonally between them in D.



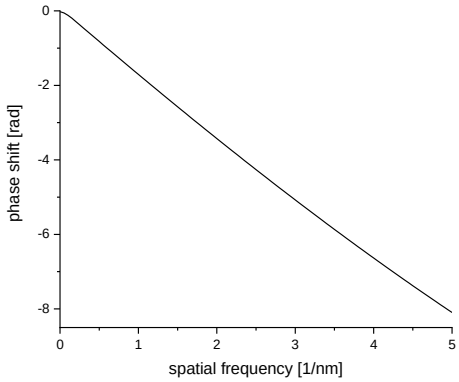
A



B



C



D

Fig. 2: A: The projected potential in V nm along a beam, B: The projected potential at 45 degrees between the two beams, C: The phase shift in radians corresponding to A, D: The phase shift corresponding to B.

The image formed at Gaussian focus with this phase plate is shown in figure 3C. Obviously the effect of correcting for the CTF by phase flipping (multiplying every second transfer band in the CTF with -1) is very minor as can be seen in figure 3D. In order to make Thon rings visible in the simulation I added some low pass filtered noise to the phantom (see figures 3A and B), which imitates the granularity of a carbon support film and/or ice containing background particles of some sort. The analytically calculated phase contrast transfer function is shown in figure 3E, whereas the PCTF determined from the image is shown in figure 3F. Clearly the somewhat special shape of the Thon “rings” has to be taken into account during CTF correction.

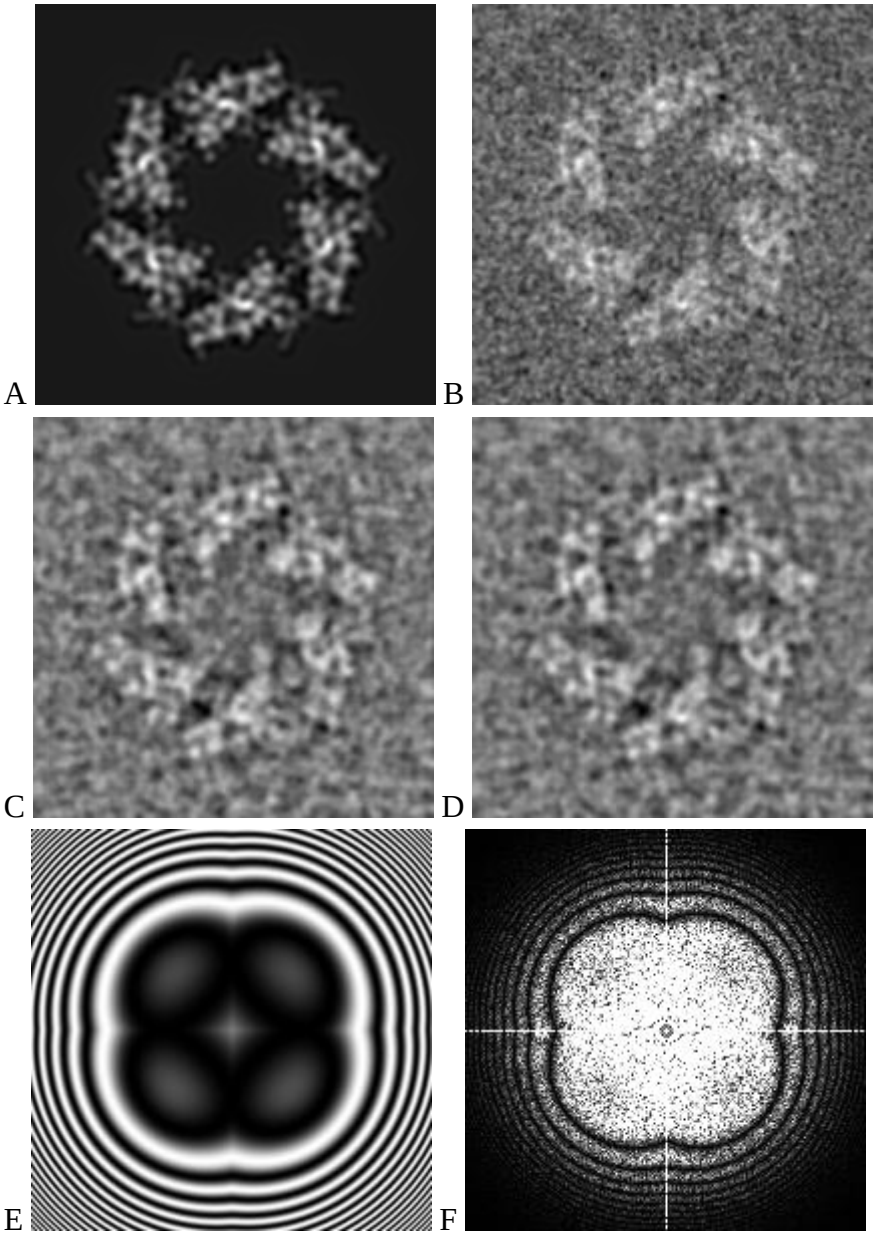
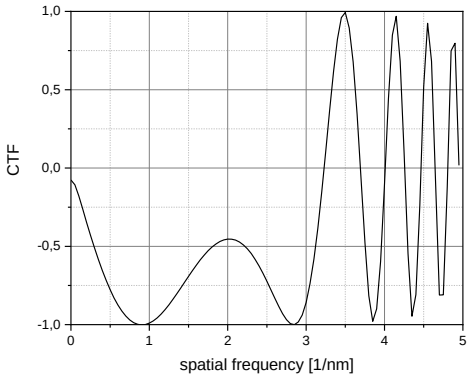
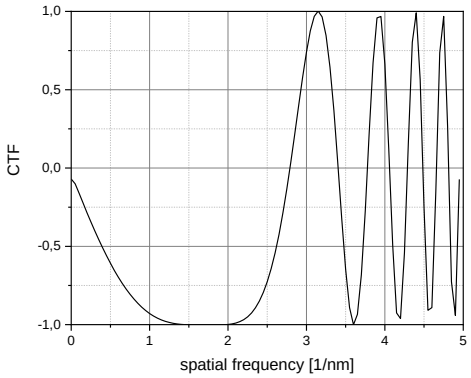


Fig. 3: A: phantom, B: phantom with added noise, C: simulated in-focus image with phase plate, D: phase flipped simulated in-focus image with phase plate, E: calculated transfer function, F: transfer function measured from image. The images (A to D) are 10 nm in size whereas the contrast transfer functions (E and F) extend to 5 nm^{-1} spatial frequency.

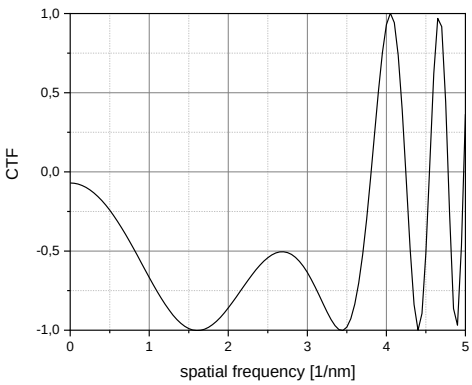
In order to evaluate the performance of the phase plate I plot the CTF with phase plate in two different directions in figure 4A and B, along a beam in A and at 45 degrees in B. For comparison I also plot the CTF without phase plate at 90 (figure 4C) and 600 nm underfocus (figure 4D). The low resolution contrast transfer in focus with phase plate is similar to that of defocusing by 600 nm without phase plate whereas the width of the pass band is similar to that achieved close to extended Scherzer defocus without the phase plate. True extended Scherzer defocus would be -86.7 nm for the chosen microscope.
In this paper I use the CTF value 0.5 as a lower limit for acceptable contrast transfer so all comparisons are based on where the absolute value of the CTF rises above 0.5.



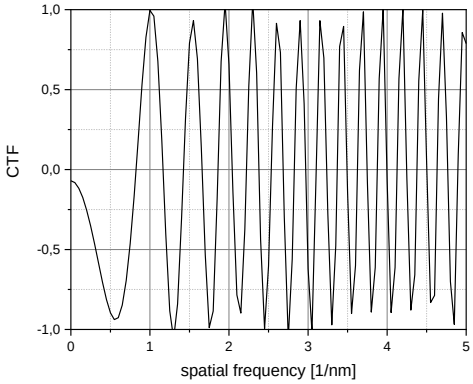
A



B



C



D

Fig. 4: A: CTF in focus with phase plate at 45 degrees, B: CTF in focus with phase plate at 0 degrees, C: CTF without phase plate at 90 nm underfocus, D: CTF without phase plate at 600 nm underfocus

The low frequency performance can be improved by underfocusing even while using the phase plate. Figure 5 shows simulated images, calculated CTFs and Thon “rings” for images recorded at 500 nm underfocus with the phase plate. A comparison of a plot of the CTF at 500 nm with phase plate with a CTF without phase plate (figure 6A and B) shows that the low

resolution performance is comparable to that achieved by underfocusing to 1300 nm without phase plate. The CTF with phase plate is almost isotropic in this case so it's not necessary to plot it in two different directions. One obvious advantage of using the phase plate with slight defocus compared to defocusing strongly without phase plate is that the CTF oscillations at high resolution are much slower.

Since the phase plate produces a relative phase shift that is roughly proportional to the spatial frequency there is no obvious upper limit for the strength of the phase plate if one isn't interested in the (extended) Scherzer-like CTF shown in figure 4A and B. Clearly one can improve the low resolution contrast transfer by increasing the charge density of the phase plate. Figure 7A shows this for a phase plate with twice the charge density (compared to the previous example) at 500 nm underfocus. The low resolution behavior is comparable to that of underfocusing by 2000 nm without phase plate (figure 7B).

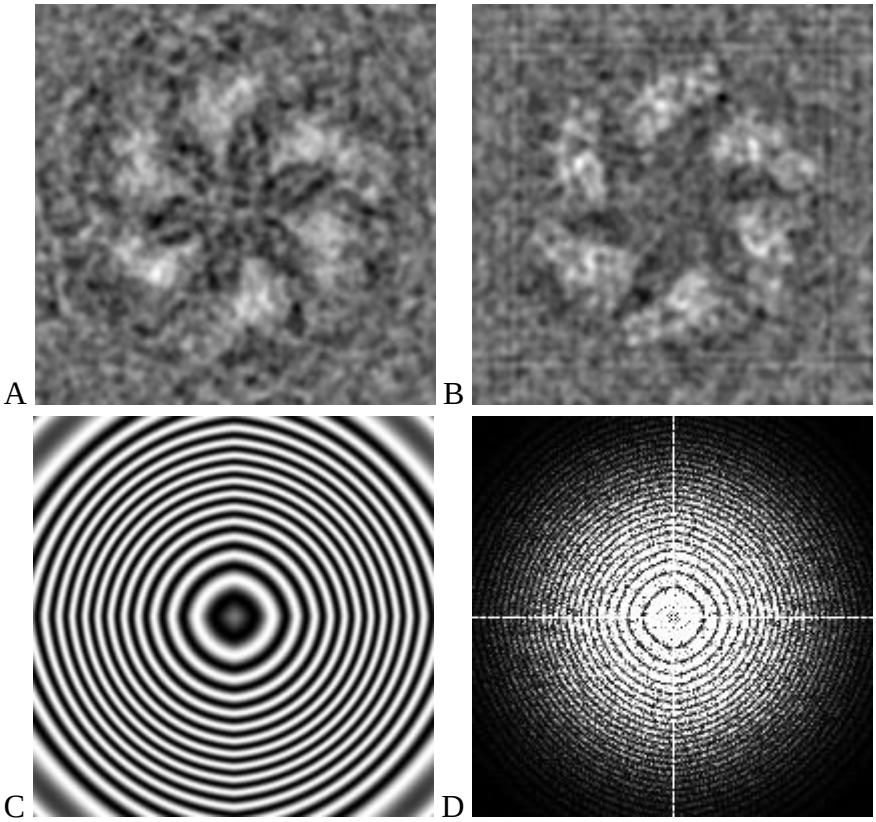
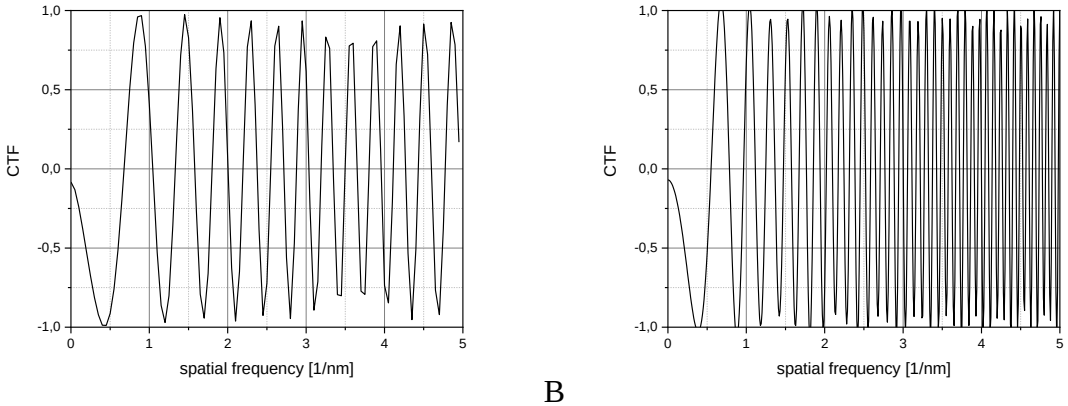
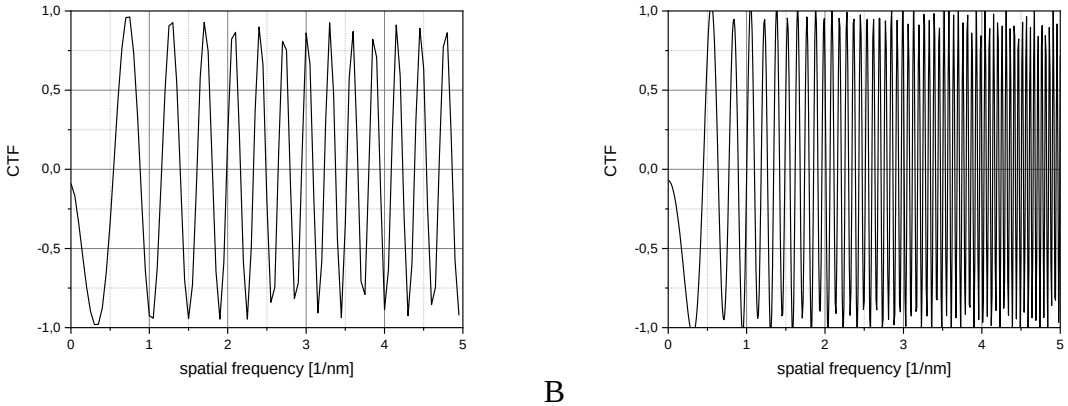


Fig. 5: A: Simulated image with phase plate at 500 nm underfocus, B: The same image after CTF correction by phase flipping, C: The calculated CTF and D: The Thon "rings"



A B

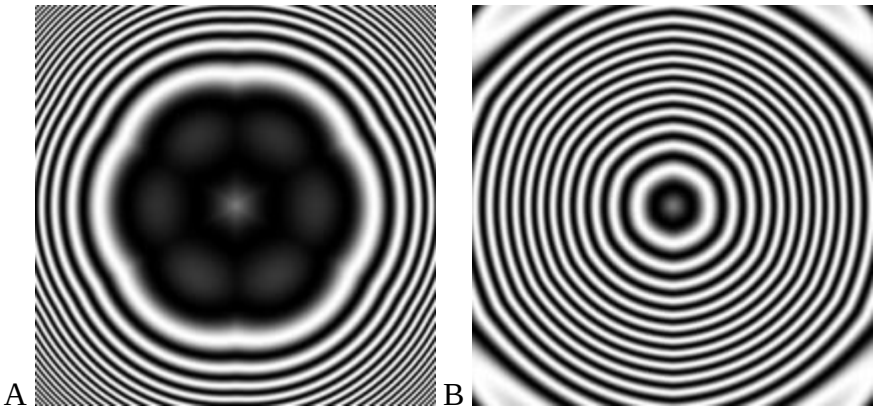
Fig. 6: A: CTF at 500 nm underfocus with phase plate (at 45 degrees), B: CTF at 1300 nm underfocus without phase plate



A B

Fig. 7: A: CTF at 500 nm underfocus with double strength phase plate (at 45 degrees), B: CTF at 2000 nm underfocus without phase plate

More than two beams, or just a single beam



A B

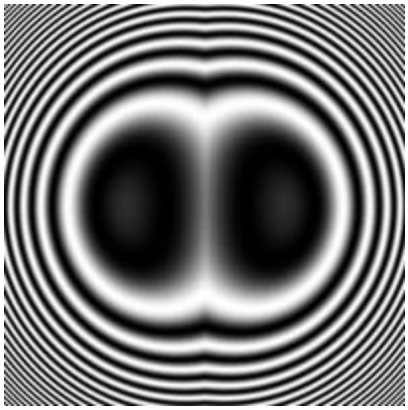
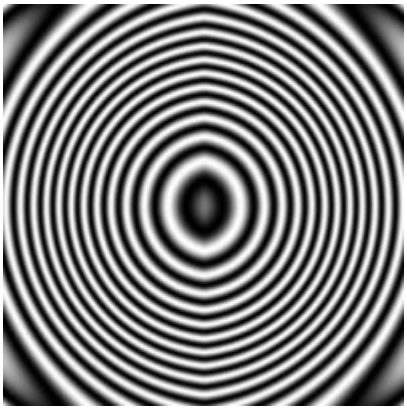
C  D 

Fig. 8: A and B: The calculated CTF in focus (A) and at 500 nm underfocus (B) with a triple beam phase plate, C and D: The calculated CTF in focus (C) and at 500 nm underfocus (D) with a single beam phase plate

Clearly one could use more than two particles beams or even just a single one. Using three or more beams has the advantage of producing more nearly circular Thon "rings" as shown in figure 8A in focus and 8B at 500 nm underfocus. However, the improvement in low resolution contrast transfer is very small and the required beam current is only about 30% smaller than for the dual beam design. The obvious disadvantage is the increased amount of hardware needed.
A single beam could also be used, but then the CTF along the beam is the same as that without phase plate. When imaging in focus, as shown in figure 8C, this means that the contrast along the beam is almost zero, whereas orthogonal to the beam an extended Scherzer-like CTF can be achieved. Underfocused images will exhibit an effect similar to astigmatism, as shown in figure 8D. The only possible advantage of using just one beam is the reduced amount of hardware. On the other hand an approximately 25 % higher beam current is necessary to achieve extended Scherzer-like contrast transfer orthogonal to the particle beam.

Conclusions

A transmission electron microscope operated close to Scherzer or extended Scherzer defocus is optimal in the sense that it has a broad band of high contrast transfer without contrast inversions and zeros, however with the disadvantage of bad contrast at low spatial frequencies. This leads to an image which is well localized and a good representation of the object, but suffers from low visibility. On the other hand, imaging at high underfocus (several hundred to several thousand nm) leads to good contrast transfer at very low spatial frequencies with the drawback of fast oscillations in the PCTF at higher resolution. This means that a strongly underfocused image of a small protein will be very delocalized and can be difficult to recognize and align in a noisy background despite its nominally high contrast. Using phase plates it is possible to improve the low resolution contrast transfer without increasing the defocus. However, all available phase plates to date have material parts that at least partially block the beam leading to problems with charging and/or additional noise.

In the present paper I discuss a design for a phase plate based on the electrostatic potential produced by beams of negatively charged particles, which contains no material parts that interact with the electron beam of the TEM apart from the charged particles themselves. In calculations and simulations I show that this phase plate adds a roughly linear term to the lens aberration function. By balancing this term with the term from spherical aberration at Gaussian focus one can achieve a Scherzer-like contrast transfer band, which is shifted to lower spatial frequencies. Alternatively, by defocusing a few hundred nm and/or increasing the charge density of the phase plate particle beam, one can achieve a PCTF with the same low-resolution transfer as that achieved by strongly defocusing without phase plate. Additionally the oscillations of the PCTF are much slower.
From what I can see, it should be possible to build this phase plate from readily available parts and add it onto a commercially available transmission electron microscope.

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